

① Bernoulli Distribution

Def:- The Bernoulli distribution is the simplest discrete probability distribution. It represents the probability distribution of a random variable that has exactly two possible outcomes: Success (with probability p) and failure (with probability $1-p$). It is used to model binary outcomes, such as a coin flip or a yes/no.

- ① Discrete Random Variable (PMF)
 - ② outcomes are Binary
- Eg :- Tossing a coin {H, T}

$$P(X=H)=0.5 \Rightarrow \text{let us consider,}$$

$$P(X=T) = 1-0.5 \Rightarrow 0.5 \Rightarrow q$$

$$p, q$$

$$q = (1-p)$$

② Whether the person will pass or fail

$$P(X=\text{pass}) = 0.4$$

$$P(X=\text{Fail}) = 1 - P(X=\text{pass})$$

Parameters

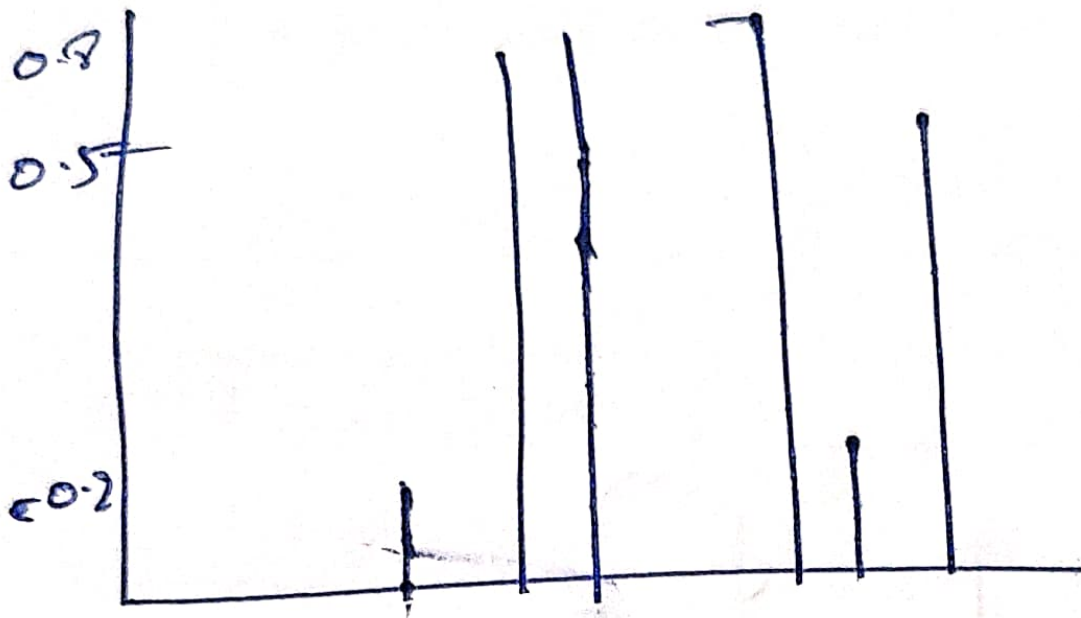
$$0 \leq p \leq 1$$

$$q = 1-p$$

$$P(X=\text{Success}) \Rightarrow K=1$$

$$P(X=\text{Fail}) = K=0$$

$$K = \{0, 1\} \Rightarrow 2 \text{ outcomes}$$



$$P(X=0) = 0.2 \text{ and } P(X=1) = 0.8$$

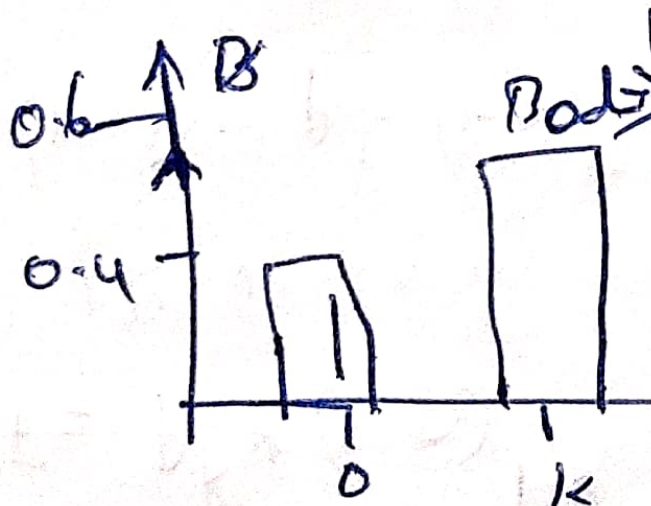
$$P(X=0) = 0.8 \text{ and } P(X=1) = 0.2$$

$$P(X=0) = 0.5 \text{ and } P(X=1) = 0.5$$

PMF :-

Company product 'A'

~~Good~~ $\Rightarrow 60\% \Rightarrow \text{Good} \Rightarrow p$



~~Bad~~ $\Rightarrow 40\% \Rightarrow \text{Bad} \Rightarrow q$

$$q = 1 - p$$

$$PMF = p^k \cdot (1-p)^{1-k}$$

$$\text{if } k=1$$

$$P(X=1) = p^1 \cdot (1-p)^{1-1}$$

$$= p$$

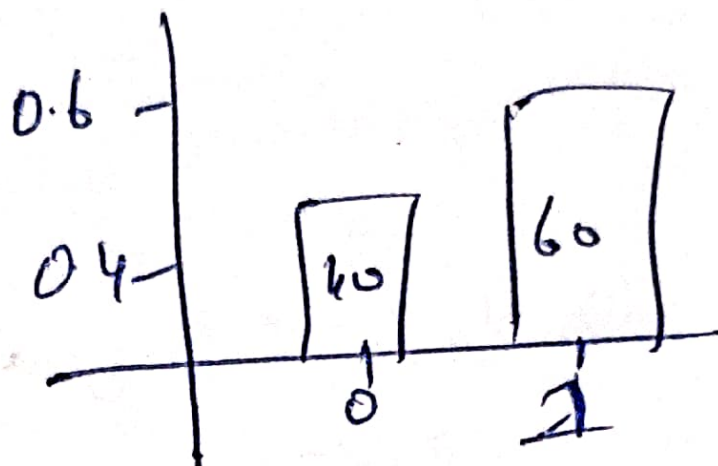
$$P(X=0) = p^0 \cdot (1-p) =$$

$$= (1-p) \Rightarrow q$$

$$PMF = \begin{cases} q = 1-p & \text{if } k=0 \\ p & \text{if } k=1 \end{cases}$$

* Mean of Bernoulli distribution

$$E(x) = \sum_{k=0}^1 k \cdot P(k)$$
$$= \sum_{k=0}^1 k \cdot P(x)$$



$$= 0 \times 0.4 + 1 \times 0.60$$

$$= 0 + 0.60$$

$$= 0.60 \Rightarrow p \text{ is Mean}$$

$p(0) \Rightarrow$ Expectation will be mean

* Median of Bernoulli distribution

$$\text{Median} \begin{cases} 0 & \text{if } p < \frac{1}{2} \\ [0, 1] & \text{if } p = \frac{1}{2} \\ 1 & \text{if } p > \frac{1}{2} \end{cases}$$

$$\text{Median} = 0 \text{ if } q > p$$

$$\text{Median} = 0.5 \text{ if } q = p$$

$$\text{Median} = 1 \text{ if } q < p$$

* Mode

$$p > q \Rightarrow p \text{ will be mode}$$

$$q > p \Rightarrow q \text{ will be mode}$$

* Variance

$$K = 0 \text{ \& \; } 1$$

$$P(X=0) =$$

$$= 0.4 (0.6)^2 + 0.6 (1-0.6)^2$$

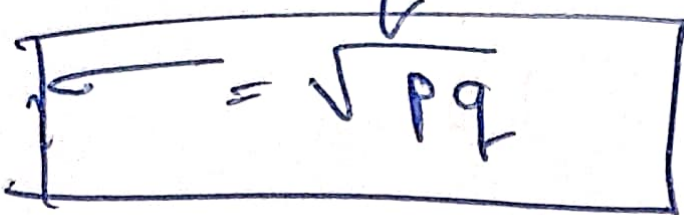
$$= 0.40 + 0.36 + 0.6 (0.16)$$

$$= 0.40 + 0.36 + 0.096$$

$$\Rightarrow P(X=0) =$$

$$P(X=1) =$$

$$q < p$$

$$\sigma = pq$$


$$\sigma = \sqrt{pq}$$